

ElectroFind — Charge & Cost Calculations

This document describes how ElectroFind estimates charging time and cost for a Hyundai Kona Electric (Long Range, 77.4 kWh usable battery).

1. The Charge Curve

The car's Battery Management System (BMS) limits how much power it will accept at each state of charge (SoC). ElectroFind encodes this as 101 data points extracted from a measured charge curve, one per integer SoC percentage (0–100%). Each point is a (SoC%, kW) pair representing the maximum power the car will accept at that SoC.

Key shape

SoC range	Accepted power	Notes
0–10%	50 → 91 kW	Ramp-up from cold
11–61%	~92–100 kW	Full-power plateau
62–67%	~75 kW	First taper
68–73%	~45 kW	Second taper
74–80%	~37–46 kW	Gradual reduction
81–82%	~25–31 kW	Sharp drop
83–92%	~25 kW	Low-power plateau
93–100%	24 → 8 kW	Final taper to full

Between integer SoC values, power is computed by **linear interpolation**:

```
powerAtSoc(soc) = curve[floor(soc)] + (curve[ceil(soc)] - curve[floor(soc)]) × (soc - floor(soc))
```

2. Charging Efficiency

Not all power drawn from the charger reaches the battery. The loss depends on whether the charger is AC or DC:

Charger type	Threshold	Efficiency	Reason
DC fast charge	≥ 22 kW	95%	The charger converts AC→DC externally. Losses are small and occur outside the car.
AC charge	< 22 kW	88%	The car's onboard charger converts AC→DC internally, generating heat (~12% lost).

Conservative estimate: costs and times are calculated using the grid draw (what the meter measures), not the energy that reaches the battery. On AC charging you pay for roughly 12% more kWh than go into the battery.

3. Time Simulation

The simulation steps through the charge in **0.1% SoC increments** from `startSoc` to `targetSoc` (or until the stay time is exhausted).

Per-step calculation

1. **Energy per step** — fixed, regardless of charger speed:

```
energyPerStep = 77.4 kWh × 0.001 = 0.0774 kWh
```

2. **Effective power** — the lower of what the charger can deliver and what the car will accept:

```
effectiveKw = min(chargerMaxKw,  
powerAtSoc(currentSoc))
```

3. **Time for this step** — divided by efficiency so AC sessions correctly take longer:

```
timeStep (minutes) = energyPerStep ÷ (effectiveKw ×  
efficiency) × 60
```

Operator precedence: $(\text{energyPerStep} \div (\text{effectiveKw} \times \text{efficiency}))$ gives hours; $\times 60$ converts to minutes.

4. **Stay-time check** — if a stay limit is set and the next step would exceed it, a partial step is applied proportional to the remaining time, then the simulation stops:

```
remaining = stayMinutes - accumulatedMinutes  
fraction = remaining ÷ timeStep  
energyAdded = energyPerStep × fraction  
socGained = 0.1% × fraction
```

Outputs

Field	Meaning
endSocPercent	SoC reached at the end of the session
energyKwh	Net energy delivered to the battery (kWh)
billedEnergyKwh	Grid draw = $\text{energyKwh} \div \text{efficiency}$ — what the meter records
chargeMinutes	Total time actively charging

`reachedTarget`

Whether the target SoC was reached before the stay limit

4. Two Simulation Modes

Mode	How it runs	What it answers
Optimal	No stay limit (<code>stayMinutes = null</code>). Runs to target SoC.	"How long will it take to reach my target, and what will it cost?"
Stay	Stay limit set. Stops at whichever comes first: target SoC or time limit.	"If I stay for N minutes, what SoC will I reach and what will it cost?"

5. Cost Calculation

Given a `SimResult` , the total cost is:

```
cost = billedEnergyKwh × pricePerKwh  
+ connectionFee  
+ chargingRatePerMin × chargeMinutes  
+ parkingRatePerMin × idleMinutes
```

Where:

```
idleMinutes = max(0, stayMinutes - chargeMinutes)
```

`idleMinutes` is the time after the car has finished charging but the session is still running (e.g. the car reached 80% after 25 minutes but you stayed for 30). Chargers that levy an idle/parking fee apply it to this window.

Tariff components sourced from Electerverse

Field	Applied to
Price per kWh	Billed grid energy
Connection fee	Fixed per session
Per-minute charging rate	Time spent actively charging
Per-minute idle/parking rate	Time in session after charging completes

Not all chargers publish all components. Where a field is absent it is treated as zero. Costs should be treated as estimates — actual billing may vary by network and tariff.

6. Limitations

- The charge curve is specific to the **Hyundai Kona Electric Long Range (77.4 kWh)**. Other vehicles will charge differently.
- The curve was measured under nominal conditions. Cold batteries charge more slowly; very hot batteries may also throttle.
- Efficiency figures (95% DC / 88% AC) are typical values. Actual efficiency varies by charger, cable, and temperature.
- The simulation does not model pre-conditioning, cell balancing pauses, or network-imposed power limits.
- Tariff data comes from Electerverse and may be incomplete or out of date.